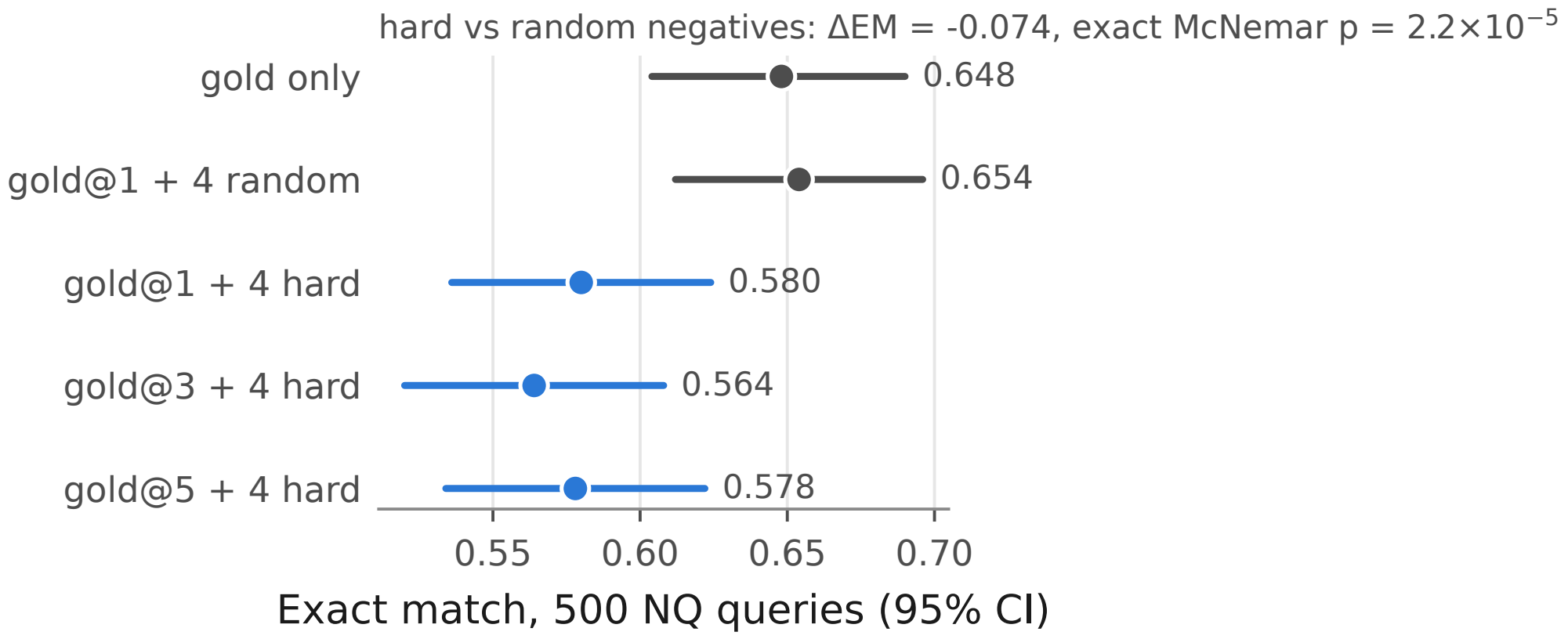
hard vs random negatives: ΔEM = -0.074, exact McNemar p = 2.2×10⁻⁵
gold only
0.648
gold@1 + 4 random
0.654
gold@1 + 4 hard
0.580
gold@3 + 4 hard
0.564
gold@5 + 4 hard
0.578
0.55
0.60
0.65
0.70
Exact match, 500 NQ queries (95% CI)